

SUBJECT 081 – PRINCIPLES OF FLIGHT – AEROPLANES

ED Decision 2020/018/R

- (1) The following standard symbols and their corresponding meanings are used for certain mathematical operations:
 - * multiplication
 - $\geq$ greater than or equal to
 - $\leq$ less than or equal to
 - SQRT(...) square root of the function, symbol or number in round brackets
- (2) Normally, it should be assumed that the effect of a variable under review is the only variation that needs to be addressed, unless specifically stated otherwise.
- (3) Candidates are expected in simple calculations to be able to convert knots (kt) into metres/second (m/s), and know the appropriate conversion factors by heart.
- (4) In the subsonic range, as covered under Subject 081 01, compressibility effects normally are not considered, unless specifically mentioned.
- (5) For those questions related to propellers (Subject 081 07), as a simplification of the physical reality, the inflow speed into the propeller plane is taken as the aeroplane's true airspeed (TAS).
- (6) In addition, when discussing propeller rotational direction, it will always be specified as seen from behind the propeller plane.
- (7) Note that the term 'mass' is used to describe a quantity of matter, and 'weight' when describing the force. However, the term 'weight' is normally used in aviation to colloquially describe 'mass'. The professional pilot should always note the units to determine whether the term 'weight' is being used to describe a force (e.g. unit newton) or quantity of matter (e.g. unit kilogram).

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
080 00 00 00		PRINCIPLES OF FLIGHT										
081 00 00 00		PRINCIPLES OF FLIGHT — AEROPLANES										
081 01 00 00		SUBSONIC AERODYNAMICS										
081 01 01 00		Basics, laws and definitions										
081 01 01 01		Laws and definitions										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		List the international system of units of measurement (SI) for mass, acceleration, weight, velocity, energy, density, temperature, pressure, force, wing loading, and power.	X	X								
(02)	X	Define ‘mass’, ‘force’, ‘acceleration’, and ‘weight’.	X	X								
(03)		State and interpret Newton’s three laws of motion.	X	X								
(04)	X	Explain air density.	X	X								
(05)	X	List the atmospheric properties that effect air density.	X	X								
(06)		Explain how temperature and pressure changes affect air density.	X	X								
(07)	X	Define ‘static pressure’.	X	X								
(08)	X	Define ‘dynamic pressure’.	X	X								
(09)	X	State the formula for ‘dynamic pressure’.	X	X								
(10)		Describe dynamic pressure in terms of an indication of the energy in the system, and how it is related to indicated airspeed (IAS) and air density for a given altitude and speed.	X	X								
(11)		State Bernoulli’s equation for incompressible flow.	X	X								
(12)		Define ‘total pressure’ and explain that the total pressure differs in different systems.	X	X								
(13)		Apply Bernoulli’s equation to flow through a venturi stream tube for incompressible flow.	X	X								
(14)		Describe how IAS is acquired from the pitot static system.	X	X								
(15)		Describe the relationship between density, temperature, and pressure for air.	X	X								

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			ATPL	CPL	ATPL/IR	ATPL	CPL					
(16)		Explain the equation of continuity and its application to the flow through a stream tube.	X	X								
(17)	X	Define 'IAS', 'CAS', 'EAS', and 'TAS'.	X	X								
081 01 01 02		Basics of airflow										
(01)	X	Describe steady and unsteady airflow.	X	X								
(02)	X	Explain the concept of a streamline and a stream tube.	X	X								
(03)	X	Describe and explain airflow through a stream tube.	X	X								
(04)	X	Explain the difference between two- and three-dimensional airflow.	X	X								
081 01 01 03		Aerodynamic forces on aerofoils										
(01)		Describe the originating point and direction of the resultant force caused by the pressure distribution around an aerofoil.	X	X								
(02)	X	Resolve the resultant force into the components 'lift' and 'drag'.	X	X								
(03)		Describe the direction of lift and drag.	X	X								
(04)	X	Define the 'aerodynamic moment'.	X	X								
(05)	X	List the factors that affect the aerodynamic moment.	X	X								
(06)		Describe the aerodynamic moment for a symmetrical aerofoil.	X	X								
(07)		Describe the aerodynamic moment for a positively and negatively cambered aerofoil.	X	X								
(08)	X	Define 'angle of attack' (α).	X	X								
081 01 01 04		Shape of an aerofoil section										
(01)	X	Describe the following parameter of an aerofoil section: leading edge.	X	X								

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			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)	X	Describe the following parameter of an aerofoil section: trailing edge.	X	X								
(03)		Describe the following parameter of an aerofoil section: chord line.	X	X								
(04)		Describe the following parameter of an aerofoil section: thickness-to-chord ratio or relative thickness.	X	X								
(05)		Describe the following parameter of an aerofoil section: location of maximum thickness.	X	X								
(06)		Describe the following parameter of an aerofoil section: camber line.	X	X								
(07)		Describe the following parameter of an aerofoil section: camber.	X	X								
(08)	X	Describe the following parameter of an aerofoil section: nose radius.	X	X								
(09)	X	Describe a symmetrical and an asymmetrical aerofoil section.	X	X								
081 01 01 05		Wing shape										
(01)	X	Describe the following parameter of a wing: span.	X	X								
(02)	X	Describe the following parameter of a wing: tip and root chord.	X	X								
(03)		Describe the following parameter of a wing: taper ratio.	X	X								
(04)	X	Describe the following parameter of a wing: wing area.	X	X								
(05)		Describe the following parameter of a wing: wing planform.	X	X								

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			ATPL	CPL	ATPL/IR	ATPL	CPL					
(06)	X	Describe the following parameter of a wing: mean geometric chord.	X	X								
(07)		Describe the following parameter of a wing: mean aerodynamic chord (MAC).	X	X								
(08)		Describe the following parameter of a wing: aspect ratio.	X	X								
(09)	X	Describe the following parameter of a wing: dihedral angle.	X	X								
(10)	X	Describe the following parameter of a wing: sweep angle.	X	X								
(11)	X	Describe the following parameter of a wing: wing twist, geometric and aerodynamic.	X	X								
(12)		Describe the following parameter of a wing: angle of incidence. <i>Remark: In certain textbooks, angle of incidence is used as angle of attack (α). For Part-FCL theoretical knowledge examination purposes, this use is discontinued, and the angle of incidence is defined as the angle between the aeroplane longitudinal axis and the wing-root chord line.</i>	X	X								
081 01 02 00		Two-dimensional airflow around an aerofoil										
081 01 02 01		Streamline pattern										
(01)	X	Describe the streamline pattern around an aerofoil.	X	X								
(02)		Describe converging and diverging streamlines, and their effect on static pressure and velocity.	X	X								
(03)	X	Describe upwash and downwash.	X	X								
081 01 02 02		Stagnation point										
(01)		Describe the stagnation point.	X	X								

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			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)		Describe the movement of the stagnation point as the α changes.	X	X								
081 01 02 03		Pressure distribution										
(01)		Describe pressure distribution and local speeds around an aerofoil including effects of camber and α .	X	X								
(02)		<i>Intentionally left blank</i>										
081 01 02 04		Centre of pressure (CP) and aerodynamic centre (AC)										
(01)		Explain CP and AC.	X	X								
081 01 02 05		<i>Intentionally left blank</i>										
081 01 02 06		Drag and wake										
(01)	X	List two physical phenomena that cause drag.	X	X								
(02)		Describe skin friction drag.	X	X								
(03)		Describe form (pressure) drag.	X	X								
(04)	X	Explain why drag and wake cause loss of energy (momentum).	X	X								
081 01 02 07		Influence of angle of attack (α)										
(01)		Explain the influence of α on lift.	X	X								
081 01 02 08		<i>Intentionally left blank</i>										
081 01 02 09		The lift coefficient (C_L) – angle of attack (α) graph										
(01)		Describe the C_L – α graph.	X	X								
(02)		Explain the significant points: — point where the curve crosses the horizontal axis (zero lift); — point where the curve crosses the vertical axis ($\alpha = 0$);	X	X								

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			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— point where the curve reaches its maximum (C_{LMAX}).										
081 01 03 00		Coefficients										
081 01 03 01		General use of coefficients										
(01)	X	Explain why coefficients are used in general.	X	X								
081 01 03 02		The lift coefficient (C_L)										
(01)		Explain the lift formula, the factors that affect lift, and perform simple calculations.	X	X								
(02)		Describe the effect of camber on the C_L - α graph (symmetrical and positively/negatively cambered aerofoils).	X	X								
(03)		Describe the typical difference in the C_L - α graph for fast and slow aerofoil design.	X	X								
(04)	X	Define ' C_{LMAX} ' (maximum lift coefficient) and ' α_{CRIT} ' (stalling α) on the graph.	X	X								
(05)		Describe C_L and explain the variables that affect it in low subsonic flight.	X	X								
081 01 03 03		Drag										
(01)		Describe the two-dimensional drag formula.	X	X								
(02)		Discuss the effect of the shape of a body, cross-sectional area, and surface roughness on the drag coefficient.	X	X								
081 01 04 00		Three-dimensional airflow around an aeroplane										
081 01 04 01		Angle of attack (α)										
(01)	X	Define 'angle of attack' (α). <i>Remark: For theoretical knowledge examination purposes, the angle-of-attack definition requires a reference line. This reference line for 3D has</i>	X	X								

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			ATPL	CPL	ATPL/IR	ATPL	CPL					
		<i>been chosen to be the longitudinal axis and for 2D the chord line.</i>										
(02)		Explain the difference between the α and the attitude of an aeroplane.	X	X								
081 01 04 02		Streamline pattern										
(01)		Describe the general streamline pattern around the wing, tail section, and fuselage.	X	X								
(02)		Explain and describe the causes of spanwise flow over top and bottom surfaces.	X	X								
(03)		Describe wing tip vortices and their contribution to downwash behind the wing.	X	X								
(04)		Explain why wing tip vortices vary with α .	X	X								
(05)		Describe spanwise lift distribution including the effect of wing planform.	X	X								
(06)		Describe the causes, distribution and duration of the wake turbulence behind an aeroplane.	X	X								
(07)		Describe the influence of flap deflection on the wing tip vortex.	X	X								
(08)		Describe the parameters that influence wake turbulence.	X	X								
081 01 04 03		Induced drag										
(01)		Explain the factors that cause induced drag.	X	X								
(02)		Describe the approximate formula for the induced drag coefficient (including variables but excluding constants).	X	X								
(03)		Describe the relationship between induced drag and total drag in straight and level flight with variable speed.	X	X								
(04)		Describe the effect of mass on induced drag at a given IAS.	X	X								

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			ATPL	CPL	ATPL/IR	ATPL	CPL					
(05)		Describe the means to reduce induced drag: — aspect ratio; — winglets; — tip tanks; — wing twist; — camber change.	X	X								
(06)		Describe the influence of lift distribution on induced drag.	X	X								
(07)		Describe the influence of downwash on the effective airflow.	X	X								
(08)		Explain induced and effective local α .	X	X								
(09)		Explain the influence of the induced α on the direction of the lift vector.	X	X								
(10)		Explain the relationship between induced drag and: — speed; — aspect ratio; — wing planform; — bank angle in a horizontal coordinated turn.	X	X								
(11)		Explain the induced drag coefficient and its relationship with the lift coefficient and aspect ratio.	X	X								
(12)		Explain the influence of induced drag on:	X	X								

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		- the C_L-α graph, and show the effect on the graph when comparing high- and low-aspect ratio wings; - the C_L-C_D (aeroplane polar), and show the effect on the graph when comparing high- and low-aspect ratio wings; - the parabolic aeroplane polar in a graph and as a formula [$C_D = C_{PD} + kC_L^2$], where C_D = coefficient of drag and C_{PD} = coefficient of parasite drag.										
(13)		Describe the C_L - C_D graph (polar).	X	X								
(14)		Indicate minimum drag on the graph.	X	X								
(15)		Explain why the C_L - C_D ratio is important as a measure of performance.	X	X								
(16)		Intentionally left blank										
081 01 05 00		Total drag										
081 01 05 01		Total drag in relation to parasite drag and induced drag										
(01)	X	State that total drag consists of parasite drag and induced drag.	X	X								
081 01 05 02		Parasite drag										
(01)		Describe the types of drag that are included in parasite drag.	X	X								
(02)		Describe form (pressure) drag and the factors which affect its magnitude.	X	X								
(03)		Describe interference drag and the factors which affect its magnitude.	X	X								
(04)		Describe friction drag and the factors which affect its magnitude.	X	X								

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			ATPL	CPL	ATPL/IR	ATPL	CPL					
081 01 05 03		Parasite drag and speed										
(01)		Describe the relationship between parasite drag and speed.	X	X								
081 01 05 04		Induced drag and speed (Refer to 081 01 04 03)										
081 01 05 05		Total drag										
(01)		Explain the total drag–speed graph and the constituent drag components.	X	X								
(02)		Indicate the speed for minimum drag.	X	X								
081 01 05 06		Intentionally left blank										
081 01 05 07		Variables affecting the total drag–speed graph										
(01)		Describe the effect of aeroplane gross mass on the graph.	X	X								
(02)		Describe the effect of pressure altitude on: — drag–IAS graph; — drag–TAS graph.	X	X								
(03)		Describe speed stability from the graph.	X	X								
(04)		Describe non-stable, neutral, and stable IAS regions.	X	X								
(05)		Explain what happens to the IAS and drag in the non-stable region if speed suddenly decreases and why this could occur.	X	X								
081 01 06 00		Ground effect										
081 01 06 01		Influence of ground effect										
(01)		Explain the influence of ground effect on wing tip vortices, downwash, airflow pattern, lift, and drag.	X	X								

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			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)		Describe the influence of ground effect on induced α and the coefficient of induced drag (C_{Di}).	X	X								
(03)		Explain the effects of entering and leaving ground effect.	X	X								
081 01 06 02		Effect on stalling angle of attack (α_{CRIT})										
(01)		Describe the influence of ground effect on α_{CRIT} .	X	X								
081 01 06 03		Effect on lift coefficient (C_L)										
(01)		Describe the influence of ground effect on the effective α and C_L .	X	X								
081 01 06 04		Effect on take-off and landing characteristics of an aeroplane										
(01)		Describe the influence of ground effect on take-off and landing characteristics and performance of an aeroplane.	X	X								
(02)		Describe the difference in take-off and landing characteristics of high- and low-wing aeroplanes.	X	X								
081 01 07 00		The relationship between lift coefficient and speed in steady, straight, and level flight										
081 01 07 01		Represented by an equation										
(01)		Explain the effect on C_L during speed increase/decrease in steady, straight, and level flight, and perform simple calculations.	X	X								
081 01 07 02		Represented by a graph										
(01)		Explain, by using a graph, the effect on speed of C_L changes at a given weight.	X	X								
081 01 08 00		Intentionally left blank										
081 01 09 00		CLMAX augmentation										

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			ATPL	CPL	ATPL/IR	ATPL	CPL					
081 01 09 01		Trailing-edge flaps and the reasons for their use in take-off and landing										
(01)		From the given relevant diagrams, describe or identify the following types of trailing-edge flaps: — split flaps; — plain flaps; — slotted flaps; — Fowler flaps.	X	X								
(02)		Describe how the wing's effective camber increases the C_L and C_D , and the reasons why this can be beneficial.	X	X								
(03)		Describe their effect on: — the location of CP; — pitching moments (due to wing CP movement); — stall speed.	X	X								
(04)		Compare their influence on the C_L - α graph: — indicate the variation in C_L at any given α ; — indicate their effect on C_{LMAX} ; — indicate their effect on critical α ; — indicate their effect on the α at a given C_L .	X	X								
(05)		Compare their influence on the C_L - C_D graph: — indicate how the $(C_L/C_D)_{MAX}$ differs from that of a clean wing.	X	X								

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			ATPL	CPL	ATPL/IR	ATPL	CPL					
(06)		Explain the influence of trailing-edge flap deflection on the glide angle.	X	X								
(07)		Describe flap asymmetry: — explain the effect on aeroplane controllability.	X	X								
(08)		Describe trailing-edge flap effect on take-off and landing: — explain the advantages of lower-nose attitudes; — explain why take-off and landing speeds/distances are reduced.	X	X								
(09)		Explain the effects of flap-setting errors, such as mis-selection and premature/late extension or retraction of flaps, on: — take-off and landing distance and speeds; — climb and descent performance; — stall buffet margins.	X	X								
081 01 09 02		Leading-edge devices and the reasons for their use in take-off and landing										
(01)		From the given relevant diagrams, describe or identify the different types of leading-edge high-lift devices: — Krueger flaps; — variable camber flaps; — slats.	X	X								
(02)		Describe the function of the slot.	X	X								

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			ATPL	CPL	ATPL/IR	ATPL	CPL					
(03)		Describe how the wing's effective camber increases with a leading-edge flap.	X	X								
(04)		Explain the effect of leading-edge flaps on the stall speed, also in comparison with trailing-edge flaps.	X	X								
(05)		Compare their influence on the C_L - α graph, compared with trailing-edge flaps and a clean wing: — indicate the effect of leading-edge devices on C_{LMAX}; — explain how the C_L curve differs from that of a clean wing; — indicate the effect of leading-edge devices on α_{CRIT}.	X	X								
(06)		Compare their influence on the C_L - C_D graph.	X	X								
(07)		Describe slat asymmetry: — describe the effect on aeroplane controllability.	X	X								
(08)		Explain the reasons for using leading-edge high-lift devices on take-off and landing: — explain the disadvantage of increased nose-up attitudes; — explain why take-off and landing speeds/distances are reduced.	X	X								
081 01 09 03		Vortex generators										
(01)		Explain the purpose of vortex generators.	X	X								
(02)		Describe the basic operating principle of vortex generators.	X	X								

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			ATPL	CPL	ATPL/IR	ATPL	CPL					
(03)		State their advantages and disadvantages.	X	X								
081 01 10 00		Means to reduce the C_L–C_D ratio										
081 01 10 01		<i>Spoilers and the reasons for their use in the different phases of flight</i>										
(01)		Describe the aerodynamic functioning of spoilers: — roll spoilers; — flight spoilers (speed brakes); — ground spoilers (lift dumpers).	X	X								
(02)		Describe the effect of spoilers on the C_L – α graph and stall speed.	X	X								
(03)		Describe the influence of spoilers on the C_L – C_D graph and lift-drag ratio.	X	X								
081 01 10 02		<i>Speed brakes and the reasons for their use in the different phases of flight</i>										
(01)		Describe speed brakes and the reasons for using them in the different phases of flight.	X	X								
(02)		State their influence on the C_L – C_D graph and lift-drag ratio.	X	X								
(03)		Explain how speed brakes increase parasite drag.	X	X								
(04)		Describe how speed brakes affect the minimum drag speed.	X	X								
(05)		Describe their effect on rate and angle of descent.	X	X								
081 01 11 00		Intentionally left blank										
081 01 12 00		Aerodynamic degradation										
081 01 12 01		<i>Ice and other contaminants</i>										
(01)		Describe the locations on an aeroplane where ice build-up will occur during flight.	X	X								

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(02)		Explain the aerodynamic effects of ice and other contaminants on: — lift (maximum C_L); — drag; — stall speed; — α_{CRIT} ; — stability and controllability.	X	X								
(03)		Explain the aerodynamic effects of icing during take-off.	X	X								
081 01 12 02		Deformation and modification of airframe, ageing aeroplanes										
(01)		Describe the effect of airframe deformation and modification of an ageing aeroplane on aeroplane performance.	X	X								
(02)		Explain the effect on boundary layer condition of an ageing aeroplane.	X	X								
081 02 00 00		HIGH-SPEED AERODYNAMICS										
081 02 01 00		Speeds										
081 02 01 01		Speed of sound										
(01)	X	Define 'speed of sound'.	X									
(02)		Explain the variation of the speed of sound with altitude.	X									
(03)		Explain the influence of temperature on the speed of sound.	X									
081 02 01 02		Mach number										
(01)		Define 'Mach number' as a function of TAS and speed of sound.	X									

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			ATPL	CPL	ATPL/IR	ATPL	CPL					
081 02 01 03		<i>Influence of temperature and altitude on Mach number</i>										
(01)		Explain the absence of change of Mach number with varying temperature at constant flight level and calibrated airspeed.	X									
(02)		Explain the relationship between Mach number, TAS and IAS during climb and descent at constant Mach number or IAS, and explain variation of lift coefficient, α , pitch and flight-path angle.	X									
(03)		Explain: — risk of exceeding the maximum operation speed (VMO) when descending at constant Mach number; — risk of exceeding the maximum operating Mach number (M_{MO}) when climbing at constant IAS; — risk of a low-speed stall at high altitude when climbing at a too low Mach number.	X									
081 02 01 04		<i>Compressibility</i>										
(01)		State that compressibility means that density can change along a streamline, and that this occurs in the high subsonic (from Mach 0.4), transonic, and supersonic flow.	X									
(02)	X	State that compressibility negatively affects the pressure gradient, leading to an overall reduction of the C_L .	X									
(03)	X	State that Mach number is a measure of compressibility.	X									

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			ATPL	CPL	ATPL/IR	ATPL	CPL					
(04)		Describe that compressibility increases low-speed stall speed and decreases α_{CRIT} .	X									
081 02 01 05		Subdivision of aerodynamic flow										
(01)	X	List the subdivision of aerodynamic flow: — subsonic flow below compressibility; — subsonic flow above compressibility; — transonic flow; — supersonic flow.	X									
(02)		Describe the characteristics of the flow regimes listed above.	X									
(03)		Explain why some transport aeroplanes cruise at Mach numbers above the critical Mach number (M_{CRIT}).	X									
081 02 02 00		Shock waves										
081 02 02 01		Definition of shock wave										
(01)	X	Define a 'shock wave'.	X									
081 02 02 02		Normal shock waves										
(01)		Describe a normal shock wave with respect to changes in: — static temperature; — static and total pressure; — velocity; — local speed of sound; — Mach number; — density.	X									

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)		Describe a normal shock wave with respect to orientation relative to the wing surface.	X									
(03)		Explain the influence of increasing Mach number on a normal shock wave, at positive lift, with respect to: — strength; — position relative to the wing; — second shock wave at the lower surface.	X									
(04)		Explain the influence of α on shock-wave intensity and shock-wave location at constant Mach number.	X									
081 02 03 00		Effects of exceeding the critical Mach number (M_{CRIT})										
081 02 03 01		Critical Mach number (M_{CRIT})										
(01)		Define ' M_{CRIT} '.	X									
(02)		Explain how a change in α , aeroplane weight, manoeuvres, and centre-of-gravity (CG) position influences M_{CRIT} .	X									
081 02 03 02		Effect on lift										
(01)		Describe the behaviour of C_L versus Mach number at constant α .	X									
(02)		Explain the consequences of exceeding M_{CRIT} with respect to C_L and C_{LMAX} .	X									
(03)		Explain the change in stall indicated airspeed (IAS) with altitude.	X									
(04)		Discuss the effect on α_{CRIT} .	X									
(05)		Explain the advantages of exceeding M_{CRIT} in aeroplanes with supercritical aerofoils with respect to:	X									

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— speed versus drag ratio; — specific range; — optimum altitude.										
081 02 03 03		Effect on drag										
(01)		Describe wave drag.	X									
(02)		Describe the behaviour of C_D versus Mach number at constant α .	X									
(03)		Explain the effect of Mach number on the C_L – C_D graph.	X									
(04)		Describe the effects and hazards of exceeding $M_{\text{DRAG DIVERGENCE}}$, namely: — drag rise; — instability; — Mach tuck; — shock stall.	X									
(05)		State the relation between M_{CRIT} and $M_{\text{DRAG DIVERGENCE}}$.	X									
081 02 03 04		Effect on pitching moment										
(01)		Discuss the effect of Mach number on the CP location.	X									
(02)		Describe the overall change in pitching moment above M_{CRIT} and explain the ‘tuck under’ or ‘Mach tuck’ effect.	X									
(03)	X	State the requirement for a Mach trim system to compensate for the effect of the CP movement and ‘tuck under’ effect.	X									

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(04)	X	Discuss the aerodynamic functioning of the Mach trim system.	X									
(05)		Discuss the corrective measures if the Mach trim fails.	X									
081 02 03 05		Effect on control effectiveness										
(01)		Discuss the effects on the effectiveness of control surfaces.	X									
081 02 04 00		Intentionally left blank										
081 02 05 00		Means to influence critical Mach number (M_{CRIT})										
081 02 05 01		Wing sweep										
(01)		Explain the influence of the angle of sweep on: — M_{CRIT}; — effective thickness/chord change or velocity component perpendicular to the quarter chord line.	X									
(02)		Describe the influence of the angle of sweepback at subsonic speed on: — C_{LMAX}; — efficiency of and requirement for high-lift devices; — pitch-up stall behaviour.	X									
(03)		Discuss the effect of wing sweepback on drag.	X									
081 02 05 02		Aerofoil shape										
(01)		Explain the use of thin aerofoils with reduced camber.	X									
(02)		Explain the main purpose of supercritical aerofoils.	X									
(03)		Intentionally left blank										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(04)		Explain the advantages and disadvantages of supercritical aerofoils for wing design.	X									
081 02 05 03		Vortex generators										
(01)		Explain the use of vortex generators as a means to avoid or restrict flow separation caused by the presence of a normal shock wave.	X									
081 03 00 00		Stall, Mach tuck, and upset prevention and recovery										
081 03 01 00		The stall										
081 03 01 01		Flow separation at increasing α										
(01)	X	Define the 'boundary layer'.	X	X								
(02)	X	Describe the thickness of a typical laminar and turbulent boundary layer.	X	X								
(03)		Describe the properties, advantages and disadvantages of the laminar boundary layer.	X	X								
(04)		Describe the properties, advantages and disadvantages of the turbulent boundary layer.	X	X								
(05)		Define the 'transition point'.	X	X								
(06)		Explain why the laminar boundary layer separates easier than the turbulent boundary layer does.	X	X								
(07)		Describe why the airflow over the aft part of a wing slows down as the α increases.	X	X								
(08)		Define the 'separation point' and describe its location as a function of α .	X	X								
(09)	X	Define α_{CRIT} .	X	X								
(10)		Describe in straight and level flight the influence of increasing the α and the phenomenon that may occur regarding:	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		- the forward stagnation point; - the pressure distribution; - the CP location (straight and swept-back wing); C_L; C_D and D (drag); - the pitching moment (straight and swept-back wing); - buffet onset; - deterrent buffet for a clean wing at high Mach number; - lack of pitch authority; - uncommanded pitch down; - uncommanded roll.										
(11)		Explain what causes the possible natural buffet on the aeroplane in a pre-stall condition.	X	X								
(12)		Describe the effectiveness of the flight controls in a pre-stall condition.	X	X								
(13)		Describe and explain the normal post-stall behaviour of a straight-wing aeroplane.	X	X								
(14)		Describe the effect and dangers of using the controls close to the stall.	X	X								
(15)		Describe the deterrent buffet.	X	X								
(16)		Explain the occurrence of the deterrent buffet and why this phenomenon is considered to be a stall limit.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
081 03 01 02		The stall speed										
(01)		Explain V_{S0} , V_{S1} , V_{SR} , and V_{S1G} .	X	X								
(02)		Solve V_{S1G} from the lift formula given varying C_L .	X	X								
(03)		Describe and explain the influence of the following parameters on stall speed: — CG; — thrust component; — slipstream; — wing loading; — mass; — wing contamination; — angle of sweep; — altitude (for compressibility effects, see 081 02 03 02).	X	X								
(04)	X	Define the 'load factor n '.	X	X								
(05)		Explain why the load factor increases in a turn.	X	X								
(06)		Explain why the load factor increases in a pull-up and decreases in a push-over manoeuvre.	X	X								
(07)		Describe and explain the influence of the 'load factor n ' on stall speed.	X	X								
(08)	X	Explain the expression 'accelerated stall'. <i>Remark: Sometimes, accelerated stall is also erroneously referred to as high-speed stall. This latter expression will not be used for Subject 081.</i>	X	X								
(09)		Calculate the change of stall speed as a function of the load factor.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(10)		Calculate the increase of stall speed in a horizontal coordinated turn as a function of bank angle.	X	X								
(11)		Calculate the change of stall speed as a function of the gross mass.	X	X								
081 03 01 03		The initial stall in spanwise direction										
(01)		Explain the initial stall sequence on the following planforms: — elliptical; — rectangular; — moderate and high taper; — sweepback or delta.	X	X								
(02)		Explain the purpose of aerodynamic and geometric twist (washout).	X	X								
(03)		<i>Intentionally left blank</i>										
(04)		Explain the influence of fences, vortilons, saw teeth, vortex generators, and strakes on engine nacelles.	X	X								
081 03 01 04		Stall warning										
(01)	X	Explain why stall warning is necessary.	X	X								
(02)	X	Explain when aerodynamic and artificial stall warnings are used.	X	X								
(03)		Explain why CS-23 and CS-25 require a margin to stall speed for take-off and landing speeds.	X	X								
(04)	X	Describe: — buffet; — stall strip;	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		- flapper switch (leading-edge stall-warning vane); - angle-of-attack vane; - angle-of-attack probe; - stick shaker.										
(05)		Describe the recovery after: - stall warning; - stall; - stick-pusher actuation.	X	X								
081 03 01 05		Special phenomena of stall										
(01)		<i>Intentionally left blank</i>										
(02)		Explain the difference between power-off and power-on stalls and recovery.	X	X								
(03)		Describe stall and recovery in a climbing and descending turn.	X	X								
(04)		Describe the pitch-up effect on a swept wing aeroplane and also an aeroplane with a T-tail.	X	X								
(05)		Describe super stall or deep stall.	X	X								
(06)		Describe the philosophy behind the stick-pusher system.	X	X								
(07)		Describe the factors that can lead to the absence of stall warning and explain the associated risks.	X	X								
(08)		Describe the indications and explain the consequences of premature stabiliser stall due to ice contamination (negative tail stall).	X	X								
(09)		Describe when to expect in-flight icing.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(10)		Explain how the effect is changed when retracting/extending lift-augmentation devices.	X	X								
(11)		<i>Intentionally left blank</i>										
(12)		Explain the effect of a contaminated wing on the stall speed and α_{CRIT} .	X	X								
(13)		Explain the hazards associated with airframe contamination when parked and during ground operations in winter conditions, and the aerodynamic effects when attempting a take-off.	X	X								
(14)		Explain de-icing/anti-icing holdover time and the likely hazards after it has expired.	X	X								
(15)		Describe the aerodynamic effects of heavy tropical rain on stall speed and drag, and the appropriate mitigation in such conditions.	X	X								
081 03 01 06		The spin										
(01)		Explain how to avoid spins.	X	X								
(02)		List the factors that cause a spin to develop.	X	X								
(03)		Describe an ‘incipient’ and ‘developed’ spin, recognition and recovery.	X	X								
(04)		Describe the differences in spin attitude with forward and aft CG.	X	X								
081 03 02 00		Buffet onset boundary										
081 03 02 01		Mach buffet										
(01)		Explain shock-induced separation, and describe its relationship with Mach buffet (high speed buffet) and Mach tuck.	X									
(02)		<i>Intentionally left blank</i>										
081 03 02 02		Buffet onset										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		Explain the concept of buffet margin, and describe the influence of the following parameters on the concept of buffet margin: — α; — Mach number; — pressure altitude; — mass; — load factor; — angle of bank; — CG location.	X									
(02)		Explain how the buffet onset boundary chart can be used to determine: — manoeuvrability; — buffet margin.	X									
(03)		Describe the consequences of exceeding M_{MO} : light buffet, buffet onset.	X									
(04)		Explain ‘aerodynamic ceiling’ and ‘coffin corner’.	X									
(05)		Explain the concept of the ‘1.3g’ buffet margin altitude.	X									
(06)		Find (using an example graph): — buffet free range; — aerodynamic ceiling at a given mass; — load factor and bank angle at which buffet occurs at a given mass, Mach number, and pressure altitude.	X									

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(07)		Explain why descent increases the buffet free range.	X									
081 03 03 00		Situations in which buffet or stall could occur										
081 03 03 01		Explain why buffet or stall occurs										
(01)		Explain why buffet or stall could occur in the following pilot-induced situations, and the methods to mitigate them: — inappropriate take-off configuration, detailing the consequences of errors associated with leading-edge devices; — steep turns; — go-around using take-off/go-around (TOGA) setting (underslung engines).	X	X								
(02)		Explain why buffet or stall could occur in the following environmental conditions at low altitude, and how to mitigate them: — thunderstorms; — wind shear and microburst; — turbulence; — wake turbulence; — icing conditions.	X	X								
(03)		Explain why buffet or stall could occur in the following environmental conditions at high altitude, and how to mitigate them: — thunderstorms in the intertropical convergence zone (ITCZ); — jet streams;	X									

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— clear-air turbulence.										
(04)		Explain why buffet or stall could occur in the following situations, and how to mitigate them: — inappropriate autopilot climb mode; — loss of, or unreliable, airspeed indication.	X	X								
081 03 04 00		Recognition of stalled condition										
081 03 04 01		Recognition and explanation of stalled condition										
(01)		Explain why a stalled condition can occur at any airspeed, or attitude or altitude.	X	X								
(02)		Explain that a stall may be recognised by continuous stall-warning activation accompanied by at least one of the following: — buffet, that can be heavy; — lack of pitch authority; — uncommanded pitch down and uncommanded roll; — inability to arrest the descent rate.	X	X								
(03)		Explain that ‘stall warning’ means a natural or synthetic indication provided when approaching the stall that may include one or more of the following indications: — aerodynamic buffeting; — reduced roll stability and aileron effectiveness; — visual or aural clues and warnings;	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		- reduced elevator (pitch) authority; - inability to maintain altitude or arrest a rate of descent; - stick-shaker activation.										
081 04 00 00		STABILITY										
081 04 01 00		Static and dynamic stability										
081 04 01 01		Basics and definitions										
(01)		Define 'static stability': - describe/identify a statically stable, neutral, and unstable condition (positive, neutral, and negative static stability), and - explain why aeroplanes are statically stable.	X	X								
(02)		Explain manoeuvrability.	X	X								
(03)		Explain the relationship between static stability and manoeuvrability.	X	X								
(04)		Define 'dynamic stability': - describe/identify a dynamically stable, neutral, and unstable motion (positive, neutral, and negative dynamic stability); - describe/identify periodic and aperiodic motion.	X	X								
(05)		<i>Intentionally left blank</i>										
081 04 01 02		Precondition for static stability										
(01)	X	Explain an equilibrium of forces and moments as the initial condition for static stability.	X	X								
081 04 01 03		Sum of forces										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)	X	Identify the forces considered in the equilibrium of forces.	X	X								
081 04 01 04		Sum of moments										
(01)		Identify the moments about all three axes considered in the equilibrium of moments.	X	X								
(02)		Discuss the effect of sum of moments not being zero.	X	X								
081 04 02 00		Intentionally left blank										
081 04 03 00		Static and dynamic longitudinal stability										
081 04 03 01		Methods for achieving balance										
(01)	X	Explain the stabiliser as the means to satisfy the condition of nullifying the total sum of the moments about the lateral axis.	X	X								
(02)		Explain the influence of the location of the wing CP relative to the CG on the magnitude and direction of the balancing force on the stabiliser.	X	X								
(03)		Explain the influence of the indicated airspeed on the magnitude and direction of the balancing force on the stabiliser.	X	X								
(04)		Explain the use of the elevator deflection or stabiliser angle for the generation of the balancing force and its direction.	X	X								
(05)		Explain the elevator deflection required to balance thrust change as a function of engine position.	X	X								
081 04 03 02		Static longitudinal stability										
(01)		Discuss the effect of the CG location on pitch manoeuvrability and longitudinal stability.	X	X								
081 04 03 03		Neutral point										
(01)	X	Define 'neutral point'.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)	X	Explain why the location of the neutral point is only dependent on the aerodynamic design of the aeroplane.	X	X								
081 04 03 04		Factors affecting neutral point										
(01)		Describe the location of the neutral point relative to the locations of the aerodynamic centre of the wing and tail.	X	X								
081 04 03 05		Location of centre of gravity (CG)										
(01)		Explain the influence of the CG location on the static longitudinal stability of the aeroplane.	X	X								
(02)		Explain the CG forward and aft limits with respect to: — longitudinal control forces; — elevator effectiveness; — stability.	X	X								
(03)		Define 'static margin'.	X	X								
081 04 03 06		The $C_m-\alpha$ graph										
(01)	X	Describe the $C_m-\alpha$ graph with respect to the relationship between the slope of the graph and static stability.	X	X								
081 04 03 07		Factors affecting the $C_m-\alpha$ graph										
(01)		Explain: — the effect on the $C_m-\alpha$ graph of a shift of CG in the forward and aft direction; — the effect on the $C_m-\alpha$ graph when the elevator is moved up or down;	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		- the effect on the $C_m-\alpha$ graph when the trim is moved; - the effect of the wing contribution; - the tail contribution.										
081 04 03 08		Intentionally left blank										
081 04 03 09		Intentionally left blank										
081 04 03 10		The stick force versus speed graph (IAS)										
(01)		Explain how a pilot perceives stable static longitudinal stick force stability regarding changes in: - speed; - altitude; - mass distribution (CG location).	X	X								
081 04 03 11		Intentionally left blank										
081 04 03 12		The manoeuvring stability/stick force per g										
(01)	X	Define the 'stick force per g', and describe that the stick force increases linearly with increase in g.	X	X								
(02)		Explain why: - the stick force per g has a prescribed minimum and maximum value; - the stick force per g decreases with pressure altitude.	X	X								
081 04 03 13		Intentionally left blank										
081 04 03 14		Factors affecting the manoeuvring stability/stick force per g										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		Explain the influence on stick force per g of: — CG location; — trim setting.	X	X								
081 04 03 15		Intentionally left blank										
081 04 03 16		Dynamic longitudinal stability										
(01)		Describe the phugoid and short-period motion in terms of period, damping, variations (if applicable) in speed, altitude, and α .	X	X								
(02)		Explain why the short-period motion is more hazardous than the phugoid.	X	X								
(03)		Describe ‘pilot-induced oscillations’.	X	X								
(04)		Explain the effect of high altitude on dynamic stability.	X	X								
(05)		Describe the influence of the CG location on the dynamic longitudinal stability of the aeroplane.	X	X								
081 04 04 00		Static directional stability										
081 04 04 01		Definition and effects of static directional stability										
(01)	X	Define ‘static directional stability’.	X	X								
(02)		Explain the effects of static directional stability being too weak or too strong.	X	X								
081 04 04 02		Sideslip angle										
(01)		Define ‘sideslip angle’.	X	X								
(02)		Identify β as the symbol used for the sideslip angle.	X	X								
081 04 04 03		Yaw-moment coefficient C_n										
(01)	X	Define the ‘yawing-moment coefficient C_n ’.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)	X	Define the relationship between C_n and β for an aeroplane with static directional stability.	X	X								
081 04 04 04		C_n - β graph										
(01)	X	Explain why: — C_n depends on β; — C_n equals zero for that β that provides static equilibrium about the aeroplane's normal axis; — if no asymmetric engine thrust, flight control or loading condition prevails, the equilibrium β equals zero.	X	X								
(02)	X	Identify how the slope of the C_n - β graph is a measure for static directional stability.	X	X								
(03)	X	Identify how the slope of the C_n - β graph is affected by altitude.	X	X								
081 04 04 05		Factors affecting static directional stability										
(01)		Describe how the following aeroplane components contribute to static directional stability: — wing; — fin; — dorsal fin; — ventral fin; — angle of sweep of the wing; — angle of sweep of the fin; — fuselage at high α;	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— strakes.										
(02)		Explain the reduction in static directional stability when the CG moves aft.	X	X								
081 04 05 00		Static lateral stability										
081 04 05 01		Definition and effects of static lateral stability										
(01)	X	Define 'static lateral stability'.	X	X								
(02)		Explain the effects of static lateral stability being too weak or too strong.	X	X								
081 04 05 02		Bank angle ϕ										
(01)	X	Define 'bank angle ϕ '.	X	X								
081 04 05 03		The roll-moment coefficient C_l										
(01)	X	Define the 'roll-moment coefficient C_l '.	X	X								
081 04 05 04		Contribution of sideslip angle (β)										
(01)		Explain how without coordination the bank angle (ϕ) creates sideslip angle (β).	X	X								
081 04 05 05		The C_l-β graph										
(01)	X	Describe the C_l - β graph.	X	X								
(02)	X	Identify the slope of the C_l - β graph as a measure for static lateral stability.	X	X								
(03)	X	Identify how the slope of the C_l - β graph is affected by altitude.	X	X								
081 04 05 06		Factors affecting static lateral stability										
(01)		Explain the contribution to the static lateral stability of: — dihedral, anhedral; — high wing, low wing; — sweep angle of the wing;	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— ventral fin; — vertical tail.										
081 04 06 00		Dynamic lateral/directional stability										
081 04 06 01		<i>Intentionally left blank</i>										
081 04 06 02		<i>Tendency to spiral dive</i>										
(01)		Explain how lateral and directional stability are coupled.	X	X								
(02)		Explain how high static directional stability and low static lateral stability may cause spiral divergence (unstable spiral dive), and under which conditions the spiral dive mode is neutral or stable.	X	X								
(03)		Describe an unstable spiral dive mode with respect to deviations in speed, bank angle, nose low-pitch attitude, and decreasing altitude.	X	X								
081 04 06 03		<i>Dutch roll</i>										
(01)		Describe Dutch roll.	X	X								
(02)		Explain: — why Dutch roll occurs when the static lateral stability is higher than static directional stability; — the conditions for a stable, neutral or unstable Dutch roll motion; — the function of the yaw damper; — the actions to be taken when the yaw damper is not available.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(03)		Describe how the asymmetric nature of shock waves on both wings, at high Mach numbers, can lead to Dutch roll.	X									
081 04 06 04		Effects of altitude on dynamic stability										
(01)		Explain that increased pressure altitude reduces dynamic lateral/directional stability.	X	X								
081 05 00 00		CONTROL										
081 05 01 00		General										
081 05 01 01		Basics — The three planes and three axes										
(01)	X	Define: — lateral axis; — longitudinal axis; — normal axis.	X	X								
(02)	X	— Define: — pitch angle; — bank angle (ϕ); — yaw angle.	X	X								
(03)		Describe the motion about the three axes.	X	X								
(04)		Name and describe the devices that control these motions.	X	X								
081 05 01 02		Camber change										
(01)		State that camber is changed by movement of a control surface and explain the effect.	X	X								
081 05 01 03		Angle-of-attack (α) change										
(01)	X	Explain the influence of local α change by movement of a control surface.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
081 05 02 00		Pitch (longitudinal) control										
081 05 02 01		Elevator/all-flying tails										
(01)		Explain the working principle of the elevator/all-flying tail and describe its function.	X	X								
081 05 02 02		Downwash effects										
(01)		Explain the effect of downwash on the tailplane α .	X	X								
(02)		<i>Intentionally left blank</i>										
081 05 02 03		<i>Intentionally left blank</i>										
081 05 02 04		Location of centre of gravity (CG)										
(01)		Explain the relationship between elevator deflection and CG location to produce a given aeroplane response.	X	X								
(02)		Explain the effect of forward CG limit on pitch control.	X	X								
081 05 02 05		Moments due to engine thrust										
(01)		Describe the effect of engine thrust on pitching moments for different engine locations.	X	X								
081 05 03 00		Yaw (directional) control										
081 05 03 01		The rudder										
(01)		Explain the working principle of the rudder and describe its function. State the relationship between rudder deflection and the moment about the normal axis. Describe the effect of sideslip on the moment about the normal axis.	X	X								
081 05 03 02		Rudder limiting										
(01)		Explain why and how rudder deflection is limited on CAT aeroplanes.	X									

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
081 05 04 00		Roll (lateral) control										
081 05 04 01		Ailerons										
(01)		Explain the functioning of ailerons.	X	X								
(02)		Describe the adverse effects of aileron deflection. (Refer to Subjects 081 05 04 04 and 081 06 01 02)	X	X								
(03)		Explain why some aeroplanes have inboard and outboard ailerons.	X	X								
(04)		State that the outboard ailerons are locked beyond a given speed to prevent: — over-control; — exceeding structural limitations; — aeroelastic phenomena (flutter, divergence and aileron reversal).	X	X								
(05)		Describe the use of aileron deflection in normal flight, flight with sideslip, crosswind landings, horizontal turns, flight with one-engine-inoperative.	X	X								
(06)	X	Define ‘roll rate’.	X	X								
(07)	X	List the factors that affect roll rate.	X	X								
(08)		Describe flaperons and aileron droop.	X	X								
081 05 04 02		Intentionally left blank										
081 05 04 03		Spoilers										
(01)		Explain how spoilers can be used to control the rolling movement in combination with or instead of the ailerons.	X	X								
081 05 04 04		Adverse yaw										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		Explain why the use of ailerons induces adverse yaw.	X	X								
081 05 04 05		Means to avoid adverse yaw										
(01)		Explain how the following reduce adverse yaw: — Frise ailerons; — differential aileron deflection; — rudder aileron cross-coupling; — roll spoilers.	X	X								
081 05 05 00		Roll/yaw interaction										
081 05 05 01		Explain roll/yaw interaction										
(01)		Explain the secondary effect of roll.	X	X								
(02)		Explain the secondary effect of yaw.	X	X								
081 05 06 00		Means to reduce control forces										
081 05 06 01		Aerodynamic balance										
(01)		Describe the purpose of aerodynamic balance.	X	X								
(02)		Describe the working principle of the horn balance.	X	X								
(03)		Describe the working principle of the internal balance.	X	X								
(04)		Describe the working principle and application of: — balance tab; — anti-balance tab; — spring tab; — servo tab.	X	X								
081 05 06 02		Artificial means										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		State the differences between fully powered controls and power-assisted controls.	X	X								
(02)		Describe power-assisted controls.	X	X								
(03)		Describe the advantages of artificial feel in fully powered control.	X	X								
081 05 07 00		Fly-by-wire (FBW)										
081 05 07 01		Control laws										
(01)		Explain which parameters may be controlled in level flight with the pitch control law.	X									
(02)		Explain the advantages of using the CG position in the FBW system.	X									
(03)		Explain what type of flight-degraded control laws may be available in case of failure.	X									
(04)		Explain what are hard and soft protections.	X									
081 05 08 00		Trimming										
081 05 08 01		Reasons to trim										
(01)		State the reasons for using trimming devices.	X	X								
(02)		Explain the difference between a trim tab and the various balance tabs.	X	X								
081 05 08 02		Trim tabs										
(01)		Describe the working principle of a trim tab including cockpit indications.	X	X								
081 05 08 03		Stabiliser trim										
(01)		Describe the working principle of a stabiliser trim including the flight deck indications.	X	X								
(02)		Explain the advantages and disadvantages of a stabiliser trim compared to a trim tab.	X	X								
(03)		Explain the relationship between CG position, take-off trim setting, and stabiliser trim position.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(04)		Explain the effect of errors in the take-off stabiliser trim setting on the rotation characteristics and stick force during take-off rotation.	X	X								
(05)		Discuss the effects of jammed and runaway stabiliser.	X	X								
(06)		Explain the consequences of a jammed stabiliser during take-off, landing, and go-around.	X	X								
081 06 00 00		LIMITATIONS										
081 06 01 00		Operating limitations										
081 06 01 01		Flutter										
(01)		Describe the phenomenon of flutter and how IAS and mass distribution affects the likelihood of flutter occurrence.	X	X								
(02)		Describe the use of mass balance to alleviate the flutter problem by adjusting the mass distribution: — wing-mounted engines on pylons; — control surface mass balance.	X	X								
(03)		Explain what is the flight envelope free of flutter.	X	X								
081 06 01 02		Intentionally left blank										
(01)		<i>Intentionally left blank</i>										
081 06 01 03		Landing gear/flap operating										
(01)		Describe the reason for flap/landing gear limitations. Define 'V _{LO} '. Define 'V _{LE} '.	X	X								
(02)		Explain why there is a difference between V _{LO} and V _{LE} in the case of some aeroplane types.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(03)		Define 'V _{FE} ' and describe flap limiting speeds.	X	X								
(04)		Describe flap design features, procedures and warnings to prevent overload.	X	X								
081 06 01 04		V_{MO}, V_{NO}, and V_{NE}										
(01)	X	Define 'V _{MO} ', 'V _{NO} ', and 'V _{NE} '.	X	X								
(02)		Explain the significance of V _{MO} , V _{NO} and V _{NE} , and the differences between these airspeeds.	X	X								
(03)		Explain the hazards of flying at speeds above V _{NE} and V _{MO} .	X	X								
081 06 01 05		M_{MO}										
(01)		Define 'M _{MO} ' and state its limiting factors.	X									
081 06 02 00		Manoeuvring envelope										
081 06 02 01		Manoeuvring-load diagram										
(01)		Describe the manoeuvring-load diagram.	X	X								
(02)		Define limit and ultimate load factor, and explain what can happen if these values are exceeded.	X	X								
(03)		Define 'V _A ', 'V _B ', 'V _C ', and 'V _D '.	X	X								
(04)		Identify and explain the varying features on the V _N diagram: — load factor 'n'; — speed scale, equivalent airspeed; — equivalent airspeed envelope; — 1g stall speed; — stall boundary (refer to 081 03 01 02).	X	X								
(05)		Describe the relationship between V _{MO} or V _{NE} and V _C .	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(06)		State all the manoeuvring load-factors limits applicable to CS-23 and CS-25 aeroplanes.	X	X								
(07)		Explain the relationship between V_A and V_S in a formula, and calculate the values.	X	X								
(08)		Explain the significance of V_A and the adverse consequences of applying full, abrupt nose-up elevator deflection when exceeding V_A .	X	X								
081 06 02 02		Factors affecting the manoeuvring-load diagram										
(01)		State the relationship of mass to load-factor limits and accelerated stall speed boundary limit.	X	X								
(02)		Calculate the change of V_A with changing mass.	X	X								
(03)		Explain why V_A loses significance at higher altitude.	X									
(04)	X	Define 'M _C ' and 'M _D '.	X									
081 06 03 00		Gust envelope										
081 06 03 01		Gust-load diagram										
(01)		Recognise a typical gust-load diagram, and state the minimum gust speeds in ft/s, m/s and kt that the aeroplane must be designed to withstand at V_B to V_C and V_D .	X	X								
(02)		Discuss considerations for the selection of V_{RA} .	X	X								
(03)		Explain the adverse effects on the aeroplane when flying in turbulence.	X	X								
081 06 03 02		Factors affecting the gust-load diagram										
(01)		Describe and explain the relationship between the gust-load factor and the following: lift-curve slope, aspect ratio, angle of sweep, altitude, wing loading, weight, wing area, equivalent airspeed (EAS), and speed of vertical gust.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		(Note: For examination purposes, the ECQB questions will not be calculation based.)										
081 07 00 00		PROPELLERS										
081 07 01 00		Conversion of engine torque to thrust										
081 07 01 01		<i>Explain conversion of aerodynamic force on a propeller blade</i>										
(01)		Explain the resolution of aerodynamic force on a propeller blade element into lift and drag or into thrust and torque.	X	X								
(02)		Describe how propeller thrust and aerodynamic torque vary with IAS.	X	X								
081 07 01 02		<i>Relevant propeller parameters</i>										
(01)		Describe the geometry of a typical propeller blade element at the reference section: — blade chord line; — propeller rotational velocity vector; — true airspeed vector; — blade angle of attack; — pitch or blade angle; — advance or helix angle. Define 'geometric pitch', 'effective pitch', and 'propeller slip'. <i>Remark: For theoretical knowledge examination purposes, the following definition is used for geometric pitch: the theoretical distance a propeller would advance in one revolution at zero blade angle of attack.</i>	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)		Describe how the terms 'fine pitch' and 'coarse pitch' can be used to express blade angle.	X	X								
081 07 01 03		Blade twist										
(01)	X	Define 'blade twist'.	X	X								
(02)		Explain why blade twist is necessary.	X	X								
081 07 01 04		Fixed pitch and variable pitch/constant speed										
(01)	X	List the different types of propellers: — fixed pitch; — adjustable pitch or variable pitch (non-governing); — variable pitch (governing)/constant speed.	X	X								
(02)		Discuss the advantages and disadvantages of fixed-pitch and constant-speed propellers.	X	X								
(03)		Discuss climb and cruise propellers.	X	X								
(04)		Explain the relationship between blade angle, blade angle of attack, and airspeed for fixed and variable pitch propellers.	X	X								
(05)		Describe and explain the forces that act on a rotating blade element in normal, feathered, windmilling, and reverse operation.	X	X								
(06)		Explain the effects of changing propeller pitch at constant IAS.	X	X								
081 07 01 05		Propeller efficiency versus speed										
(01)		Define 'propeller efficiency'.	X	X								
(02)		Explain and describe the relationship between propeller efficiency and speed (TAS) for different types of propellers.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(03)		Explain the relationship between blade angle and thrust.	X	X								
081 07 01 06		Effects of ice on propeller										
(01)		Describe the effects and hazards of ice on a propeller.	X	X								
081 07 02 00		Engine failure										
081 07 02 01		Windmilling drag										
(01)		Describe the effects of an inoperative engine on the performance and controllability of an aeroplane: — thrust loss/drag increase; — influence on yaw moment during asymmetric power.	X	X								
081 07 02 02		Feathering										
(01)		Explain the reasons for feathering a propeller, including the effect on the yaw moment, performance and controllability.	X	X								
081 07 03 00		Design features for power absorption										
081 07 03 01		Propeller design characteristics that increase power absorption										
(01)	X	Name the propeller design characteristics that increase power absorption.	X	X								
081 07 03 02		Diameter of propeller										
(01)		Explain the reasons for restricting propeller diameter.	X	X								
081 07 03 03		Number of blades										
(01)	X	Define 'solidity'.	X	X								
(02)		Describe the advantages and disadvantages of increasing the number of blades.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
081 07 03 04		Propeller noise										
(01)	X	Describe how propeller noise can be minimised.	X	X								
081 07 04 00		Secondary effects of propellers										
081 07 04 01		Torque reaction										
(01)		Describe the effects of engine/propeller torque.	X	X								
(02)		Describe the following methods for counteracting engine/propeller torque: — counter-rotating propellers; — contra-rotating propellers.	X	X								
081 07 04 02		Gyroscopic precession										
(01)	X	Describe what causes gyroscopic precession.	X	X								
(02)	X	Describe the effect on the aeroplane due to the gyroscopic effect.	X	X								
081 07 04 03		Slipstream effect										
(01)		Describe the possible effects of the rotating propeller slipstream.	X	X								
081 07 04 04		Asymmetric blade effect										
(01)		Explain the asymmetric blade effect (also called P factor).	X	X								
(02)		Explain the influence of direction of rotation on the critical engine on twin-engine aeroplanes.	X	X								
081 07 04 05		Consideration of propeller effects										
(01)		Describe, given direction of propeller rotation, the propeller effects during take-off run, rotation and initial climb, and their consequence on controllability.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)		Describe, given the direction of propeller rotation, the propeller effects during a go-around and their consequence on controllability.	X	X								
(03)		Explain how propeller effects during go-around can be affected by: — high engine performance conditions and their effect on the VMC speeds; — loss of the critical engine; — crosswind; — high flap setting;	X	X								
081 08 00 00		FLIGHT MECHANICS										
081 08 01 00		Forces acting on an aeroplane										
081 08 01 01		<i>Straight, horizontal, steady flight</i>										
(01)	X	Describe the forces that act on an aeroplane in straight, horizontal, and steady flight.	X	X								
(02)	X	List the four forces and state where they act on.	X	X								
(03)		Explain how the four forces are balanced, including the function of the tailplane.	X	X								
081 08 01 02		<i>Straight, steady climb</i>										
(01)	X	Define ‘flight-path angle’ (γ).	X	X								
(02)		Describe the relationship between pitch attitude, γ and α for zero-wind and zero-bank conditions.	X	X								
(03)	X	Describe the forces that act on an aeroplane in a straight, steady climb.	X	X								
(04)		Name the forces parallel and perpendicular to the direction of flight. — Apply the formula relating to the parallel forces ($T = D + W \sin \gamma$).	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— Apply the formula relating to the perpendicular forces ($L = W \cos \gamma$).										
(05)		Explain why thrust is greater than drag.	X	X								
(06)		Explain why lift is less than weight.	X	X								
(07)		Explain the formula (for small angles) that gives the relationship between γ , thrust, weight, and lift-drag ratio, and use this formula for simple calculations.	X	X								
(08)		Explain how IAS, α , and γ change in a climb performed with constant vertical speed and constant thrust setting.	X	X								
081 08 01 03		<i>Straight, steady descent</i>										
(01)	X	Describe the forces that act on an aeroplane in a straight, steady descent.	X	X								
(02)		Name the forces parallel and perpendicular to the direction of flight. — Apply the formula for forces parallel to the direction of flight ($T = D - W \sin \gamma$). — Apply the formula relating to the perpendicular forces ($L = W \cos \gamma$).	X	X								
(03)		Explain why lift is less than weight.	X	X								
(04)		Explain why thrust is less than drag.	X	X								
081 08 01 04		<i>Straight, steady glide</i>										
(01)	X	Describe the forces that act on an aeroplane in a straight, steady glide.	X	X								
(02)		Name the forces parallel and perpendicular to the direction of flight.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		- Apply the formula for forces parallel to the direction of flight ($D = W \sin \gamma$). - Apply the formula for forces perpendicular to the direction of flight ($L = W \cos \gamma$).										
(03)		Describe the relationship between the glide gradient and the lift–drag ratio, and calculate glide range given: - initial height; - L–D ratio; - glide speed and wind speed.	X	X								
(04)		Define V_{MD} (speed for minimum drag) and explain the relationship between α , V_{MD} and the best lift–drag ratio.	X	X								
(05)		Explain the effect of wind component on glide angle, duration, and distance.	X	X								
(06)		Explain the effect of mass change on glide angle, duration, and distance, given that the aeroplane remains at either the same airspeed or at V_{MD} .	X	X								
(07)		Explain the effect of configuration change on glide angle and duration.	X	X								
(08)		Describe the relation between TAS, gradient of descent, and rate of descent.	X	X								
(09)		Define V_{MP} (speed for minimum power) and describe that the minimum rate of descent in the glide will be at V_{MP} , and explain the relationship of this speed to the optimum speed for minimum glide angle.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(10)		Discuss when a pilot could elect to fly for minimum glide rate of descent or minimum glide angle, and why speed stability or headwinds/tailwinds may favour a speed that is faster or slower than the optimum airspeed in still air.	X	X								
081 08 01 05		Steady, coordinated turn										
(01)		Describe the forces that act on an aeroplane in a steady, coordinated turn.	X	X								
(02)		Resolve the forces that act horizontally and vertically during a coordinated turn ($\tan \phi = \frac{V^2}{gR}$).	X	X								
(03)		Describe the difference between a coordinated and an uncoordinated turn, and describe how to correct an uncoordinated turn using turn and slip indicator or turn coordinator.	X	X								
(04)		Explain why the angle of bank is independent of mass, and that it only depends on TAS and radius of turn.	X	X								
(05)		Resolve the forces to show that for a given angle of bank the radius of turn is determined solely by airspeed ($\tan \phi = \frac{V^2}{gR}$).	X	X								
(06)		Calculate the turn radius of a steady turn given TAS and angle of bank.	X	X								
(07)		Explain the effects of bank angle on: – load factor ($LF = 1/\cos \phi$);	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		– α; – thrust; – drag.										
(08)	X	Define 'angular velocity'.	X	X								
(09)	X	Define 'rate of turn' and 'rate-1 turn'.	X	X								
(10)		Explain the influence of TAS on rate of turn at a given bank angle.	X	X								
(11)		Calculate the load factor and stall speed in a turn given angle of bank and 1g stall speed.	X	X								
(12)		Explain situations in which turn radius is relevant for safety, such as maximum speed limits on departure or arrival plates, or outbound speed categories on approach plates, and the implications/hazards of exceeding given speeds.	X	X								
(13)		Describe the hazards of excessive use of rudder to increase the rate of turn in a swept-wing aeroplane.	X	X								
081 08 02 00		Asymmetric thrust										
081 08 02 01		Jet-engined and propeller-driven aeroplanes										
(01)		Describe the effects on the aeroplane of asymmetric thrust during flight, for both jet-engined and propeller-driven aeroplanes.	X	X								
(02)		Explain critical engine, and explain, for a propeller-driven aeroplane, the effect of the direction of propeller rotation.	X	X								
(03)	X	Explain the effect of steady, asymmetric flight on a conventional (ball) slip indicator/turn indicator.	X	X								
(04)		Explain the effect of a crosswind on asymmetric flight.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
081 08 02 02		Balanced moments about the normal axis										
(01)		Explain the yaw moments about the CG.	X	X								
(02)		Explain the change to the yaw moment caused by the effect of air density on thrust.	X	X								
(03)		Describe the changes to the yaw moment caused by engine distance from CG.	X	X								
(04)		Describe the methods to achieve directional balance following engine loss.	X	X								
081 08 02 03		Forces parallel to the lateral axis										
(01)		Explain: — the force on the vertical fin; — the fuselage side force due to sideslip (using wing-level method); — the use of bank angle to tilt the lift vector (in wing-down method).	X	X								
(02)		Explain the flight hazards at VMC: — α ; — side slip; — loads on the fin; — α on the fin.	X	X								
(03)		Explain the effect on fin α due to sideslip.	X	X								
081 08 02 04		Influence of aeroplane mass										
(01)		Explain why controllability with one-engine-inoperative is a typical problem arising from the low speeds associated with low aeroplane mass.	X	X								
081 08 02 05		Intentionally left blank										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
081 08 02 06		Intentionally left blank										
081 08 02 07		Intentionally left blank										
081 08 02 08		Minimum control speed (V_{MC})										
(01)		Define ' V_{MC} '.	X	X								
(02)		Describe how V_{MC} is determined.	X	X								
(03)		Explain the influence of the CG location.	X	X								
081 08 02 09		Minimum control speed during approach and landing (V_{MCL})										
(01)		Define ' V_{MCL} '.	X	X								
(02)		Describe how V_{MCL} is determined.	X	X								
(03)		Explain the influence of the CG location.	X	X								
081 08 02 10		Minimum control speed on the ground (V_{MCG})										
(01)		Define ' V_{MCG} '.	X	X								
(02)		Describe how V_{MCG} is determined.	X	X								
(03)		Explain the influence of the CG location.	X	X								
081 08 02 11		Influence of density										
(01)		Describe the influence of density on thrust during asymmetric flight.	X	X								
(02)		Explain why V_{MC} , V_{MCL} and V_{MCG} reduce with a reduction in thrust.	X	X								
081 08 03 00		Significant points on a polar curve										
081 08 03 01		Identify and explain										
(01)		Identify and explain the significant points on a polar curve.	X	X								